

The coloured glass can be obtained by the addition of colouring ions. Cobalt oxide imparts deep blue colour to the glass. While cupric oxide and nickel oxide imparts black colour and ferric oxide imparts bluish-green glass.

58. With reference to ionic compounds, consider the following statements :

1. Ionic compounds are insoluble in alcohol.
2. Ionic compounds in the solid state are a good conductor of electricity.

Which of these statements is/are correct?

- (a) Only 1                                      (b) Only 2  
(c) Both 1 and 2                              (d) Neither 1 nor 2

I.A.S. (Pre) 2003

Ans. (a)

The ionic compounds are the forms of negative and positive ions so they are very stable. Ionic compounds are generally insoluble in organic solvents (like alcohol). Non-polar organic solvents cannot effectively interact with the ions and break the ionic bonds. Alcohol has very low dielectric constant due to which it would not be able to weaken the force of attraction between the ions of ionic compounds. So ionic compounds are insoluble in alcohol. For example – NaCl, KCl. Ionic compounds in the solid state are bad conductors of electricity because of absence of free electrons.

## Organic Chemistry

### A. Hydrocarbons

#### Notes

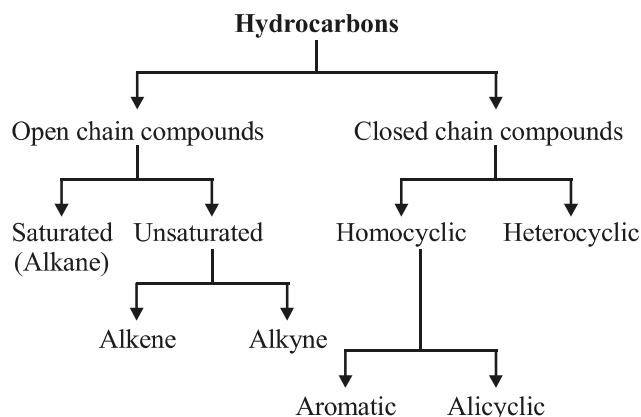
- In organic chemistry, a hydrocarbon is an organic compound consisting entirely of **hydrogen** and **carbon**.
- Hydrocarbons from which one hydrogen atom has been removed are functional groups called **hydrocarbyls**, e.g. **Methyl, phenyl** etc.
- In hydrocarbons the carbon atoms join together to form the framework of the compound and the hydrogen atoms attach to them in many different configurations.
- Hydrocarbons are the principal constituents of **petroleum** and **coal gas**.
- Hydrocarbons serve as fuels and lubricants as well as raw materials for the production of plastics, fibres, rubbers, solvents, explosives and industrial chemicals.

- Many hydrocarbons occur in nature. In addition to making up fossil fuels, they are present in trees and plants. For example in the form of pigments called **carotene** ( $C_{40}H_x$ ) that occur in **carrots** and **green leaves**.
- **Petroleum** (rock oil) is the main primary substance used to prepare many commercial substances such as compressed natural gas (CNG), liquified petroleum gas (LPG), domestic fuels and automobile fuels.
- Besides above substances, diesel, kerosene, mobil, paraffin waxes are also obtained by the fractional distillation of petroleum.

#### Classification of Hydrocarbons :

- On the basis of structure, hydrocarbons can be broadly classified into two classes :

- I. Open chain compounds
- II. Cyclic or closed chain compounds



#### I. Open chain compounds :

- The hydrocarbons, which contain atoms linked on the open-chain are known as open-chain compounds. In these hydrocarbons, the chain of carbon atoms is not cyclic. All aliphatic hydrocarbons contain open chains.
- Open chain compounds are further divided into
  - (i) Saturated hydrocarbons
  - (ii) Unsaturated hydrocarbons
- **(i) Saturated hydrocarbons (Alkanes):**
  - Those hydrocarbons, in which the carbon atoms are bonded to each other with a single **covalent bond** ( $-C-C-$ ) are called saturated hydrocarbons. They are also known as **alkanes**.
  - They follow the general formula  $C_nH_{2n+2}$  (+ane suffix). For example-methane, ethane, propane, etc.
  - They are also known as **paraffin** because these are less reactive in normal conditions with acids and other reagents (Parum = less, Affinis = reactivity).

### Methane (CH<sub>4</sub>) :

- **Methane** is the first member of open chain saturated hydrocarbons.
- Methane is the primary constituent of natural gas and it is released into the atmosphere from crude oil production and some industrial activities. More of it is released from biological processes. Soil bacteria and bacteria in guts of **termites** and **ruminants**, like cows, produce methane. Other mammals including human produce it in small amounts.
- It is also known as **Marsh gas** due to its production in the wetland by decomposition of organic materials by bacteria.

#### Some Alkanes and their Molecular Formula

| Molecular Formula               | Name    |
|---------------------------------|---------|
| CH <sub>4</sub>                 | Methane |
| C <sub>2</sub> H <sub>6</sub>   | Ethane  |
| C <sub>3</sub> H <sub>8</sub>   | Propane |
| C <sub>4</sub> H <sub>10</sub>  | Butane  |
| C <sub>5</sub> H <sub>12</sub>  | Pentane |
| C <sub>6</sub> H <sub>14</sub>  | Hexane  |
| C <sub>7</sub> H <sub>16</sub>  | Heptane |
| C <sub>8</sub> H <sub>18</sub>  | Octane  |
| C <sub>9</sub> H <sub>20</sub>  | Nonane  |
| C <sub>10</sub> H <sub>22</sub> | Decane  |
| C <sub>20</sub> H <sub>42</sub> | Icosane |

### Properties of Alkanes :

- These are non-polar.
- Weak Van der Waals force exists between bonds.
- Due to weak forces, the first four members (from C<sub>1</sub> to C<sub>4</sub>) are gases, members from C<sub>5</sub> to C<sub>17</sub> are liquids and members C<sub>18</sub> and onwards are solids at 298K (25°C).
- The alkanes are generally non-reactive with acids, bases and oxidising or reducing reagents.

### (ii) Unsaturated Hydrocarbons :

- Hydrocarbons, in which the carbon atoms are bonded to each other with two (= C = C =) or three (– C ≡ C –) covalent bonds and have a tendency to gain more hydrogen atoms are known as unsaturated hydrocarbons. They are of two types –

#### (a) Alkenes :

- Hydrocarbons containing a double bond between carbon atoms are known as alkenes.
- They have the general formula C<sub>n</sub>H<sub>2n</sub> (+ene suffix).
- They are also known as **olefins**.

### Name and Molecular Formula of Some Alkenes

| Molecular Formula             | Name                |
|-------------------------------|---------------------|
| C <sub>2</sub> H <sub>4</sub> | Ethylene (ethene)   |
| C <sub>3</sub> H <sub>6</sub> | Propylene (propene) |
| C <sub>4</sub> H <sub>8</sub> | Butylene (butene)   |

### Ethylene (C<sub>2</sub>H<sub>4</sub>) :

- It is prepared by heating ethyl alcohol at 170°C in presence of conc-sulphuric acid.
- Sulphuric acid acts as a dehydrating agent in this reaction.
$$\text{C}_2\text{H}_5\text{OH} \xrightarrow[170^\circ\text{C}]{\text{H}_2\text{SO}_4(\text{Conc.})} \text{C}_2\text{H}_4 + \text{H}_2\text{O}$$
- Ethylene is colourless, sweet smell and less soluble gas in water, on smelling unconsciousness is resulted.
- Its atomic weight is 28.
- Its density is 1.25 gm/litre at N.T.P.
- Its boiling point is –105°C.

### Uses of Ethylene –

- Some plants produce ethylene which helps in the ripening of fruits (as a natural plant hormone).
- It is used on a commercial basis for ripening of fruits.
- It is used as an anaesthetic.
- It is widely used to manufacture polythene.
- It is used in the preparation of mustard gas.
- Mustard gas is a warfare agent known as the **sulphur mustard** which can form blisters on exposed skin and in the lungs.

### (b) Alkynes :

- Hydrocarbons containing a triple bond between carbon atoms are known as an **alkyne**.
- They have the general formula C<sub>n</sub>H<sub>2n-2</sub>.
- Its first member is **acetylene** or **ethyne** (C<sub>2</sub>H<sub>2</sub>).

### Acetylene (C<sub>2</sub>H<sub>2</sub>) :

- When calcium carbide reacts with water, acetylene is formed.



- It is colourless, odourless and less soluble gas in water.
- Its boiling point is –84°C.
- Impure acetylene has smelt like **garlic**.
- It is lighter than air.
- Compressed and liquified acetylene is a dangerous explosive.
- Its mixture with oxygen on ignition explodes violently.

### Uses of Acetylene –

- When carbide comes in contact with moisture, it produces acetylene gas. Acetylene acts like ethylene and accelerates the ripening process, but it is inadvisable because calcium

carbide has carcinogenic (cancer-causing) properties. The use of this chemical for ripening fruits is illegal in most countries.

- Carbide lamps or acetylene gas lamps are simple lamps that produce and burn acetylene which is created by the reaction of calcium carbide. This lamp is used in mines.
- Oxyacetylene flame is used for welding and cutting of metals.

## II. Cyclic or closed chain compounds :

- Those hydrocarbons, which contain the ring of carbon atoms are known as cyclic ring compounds.
- They are further classified into homocyclic and heterocyclic compounds.

### (i) Homocyclic compounds :

- If the cyclic organic compounds are composed of only carbon atoms, they are known as homo-cyclic hydrocarbons.
- These are divided into—

(a) **Aromatic hydrocarbon** : Aromatic hydrocarbons are defined as those cyclic hydrocarbons which have alternate C – C and C = C bond and are similar to the benzene ring in structure. For examples - benzene, naphthalene etc.

(b) **Alicyclic hydrocarbons** : Those cyclic hydrocarbons which have no alternate C – C and C = C bond and similar properties to aliphatic compounds are known as alicyclic compounds. For example - cyclohexane, cyclopentane etc.

### (ii) Heterocyclic compounds :

- Those cyclic hydrocarbons, which are composed of carbon atoms as well as other elements like N, O, S etc. are known as heterocyclic compounds. For example—Furan and Pyridine.

## Question Bank

1. Which one of the following is the correct sequential order in which these appeared under simulated primitive Earth condition?

- Methane, Hydrogen Cyanide, Nitrile, Amino Acid
- Hydrogen Cyanide, Methane, Nitrile, Amino Acid
- Amino Acid, Nitrile, Hydrogen Cyanide, Methane
- Nitrile, Amino Acid, Methane, Hydrogen Cyanide

I.A.S. (Pre) 1994

Ans. (a)

The correct sequence of their appearance under simulated primitive Earth condition is Methane, Hydrogen Cyanide, Nitrile, Amino Acid.

2. Which is the fundamental element of all organic compounds?

- Nitrogen
- Oxygen
- Carbon
- Brimstone

56<sup>th</sup> to 59<sup>th</sup> B.P.S.C. (Pre) 2015  
U.P.U.D.A./L.D.A. (Pre) 2001

Ans. (c)

Carbon is the fundamental element of all organic compounds (including bio-compounds). The organic compounds are a large class of chemical compounds in which one or more atoms of carbon are covalently linked to atoms of other elements, most commonly hydrogen, oxygen or nitrogen.

3. In all bio-compounds the most necessary fundamental element is –

- Sulphur
- Carbon
- Oxygen
- Nitrogen

U.P.P.C.S. (Mains) 2015

Ans. (b)

See the explanation of above question.

4. Which one of the following sets of elements was primarily responsible for the origin of life on the Earth?

- Hydrogen, Oxygen, Sodium
- Carbon, Hydrogen, Nitrogen
- Oxygen, Calcium, Phosphorus
- Carbon, Hydrogen, Potassium

I.A.S. (Pre) 2012

Ans. (b)

Biologists consider that six elements were primarily responsible for the origin of life on the Earth. They are Hydrogen, Oxygen, Carbon, Nitrogen, Phosphorus and Sulphur. The matter within every living earthly creature mainly consists of the compounds these chemical elements e.g. proteins, nucleic acid, carbohydrates, vitamins, hormones, etc.

5. Which one of the following is the correct sequence in increasing order of molecular weights of the hydrocarbons?

- Methane, Ethane, Propane and Butane
- Propane, Butane, Ethane and Methane
- Butane, Ethane, Propane and Methane
- Butane, Propane, Ethane and Methane

I.A.S. (Pre) 2001

Ans. (a)

The increasing order of molecular weights of hydrocarbons is :  
Methane -  $\text{CH}_4$ , Ethane -  $\text{C}_2\text{H}_6$ , Propane -  $\text{C}_3\text{H}_8$ , Butane -  $\text{C}_4\text{H}_{10}$

6. A hydrocarbon in which two carbon atoms are joined by a double bond is called as an :

- (a) Alkane (b) Alkene  
(c) Alkyne (d) Ionic bond  
(e) None of the above/More than one of the above

63<sup>rd</sup> B.P.S.C. (Pre) 2017

Ans. (b)

A hydrocarbon in which two carbon atoms are joined by a double bond is called as an alkene. Alkenes have the general formula  $C_nH_{2n}$ . They are also known as olefins. Ethylene ( $C_2H_4$ ), Propylene ( $C_3H_6$ ) and Butylene ( $C_4H_8$ ) are first three members of this group.

7. Arrange the following carbon compounds in the decreasing order of their melting points :

- (a) Ethanol > Chloroform > Acetic acid > Methane  
(b) Chloroform > Acetic acid > Ethanol > Methane  
(c) Acetic acid > Chloroform > Methane > Ethanol  
(d) Acetic acid > Chloroform > Ethanol > Methane

Chhattisgarh P.C.S. (Pre) 2022

Ans. (d)

The arrangement of given carbon compounds in the decreasing order of their melting points is as follows :  
Acetic acid (about  $16.6^\circ\text{C}$ ) > Chloroform (about  $-63.5^\circ\text{C}$ ) > Ethanol (about  $-114.1^\circ\text{C}$ ) > Methane (about  $-182.5^\circ\text{C}$ )  
Hence, option (d) is the correct answer.

8. Which of the following gases is used in cigarette lighters?

- (a) Butane (b) Methane  
(c) Propane (d) Radon

56<sup>th</sup> to 59<sup>th</sup> B.P.S.C. (Pre) 2015

Ans. (a)

Butane is a flammable hydrocarbon with the molecular formula  $C_4H_{10}$ . It is a natural gas perhaps best known for its use as a fuel cigarette lighters. It is also an organic compound known as NGL, a Natural Gas Liquid.

9. Leakage of which of the following gases had caused the Bhopal Gas Tragedy?

- (a) Methyl isocyanate  
(b) Carbon monoxide  
(c) Nitric oxide  
(d) Sulphur dioxide

Uttarakhand P.C.S. (Pre) 2012

Ans. (a)

The Bhopal disaster or the Bhopal Gas Tragedy was a disaster that resulted from an accident. It happened at a Union Carbide subsidiary pesticide plant in Bhopal, India. On the night of 2-3 December 1984, the plant released 42 tonnes of toxic methyl isocyanate (MIC) gas, exposing more than 500,000 people to toxic gases.

10. Bhopal Gas Tragedy was caused due to the leakage of :

- (a) Methyl isocyanate (b) Nitrogen dioxide  
(c) Sulphur dioxide (d) Carbon monoxide

U.P.P.C.S. (Pre) 2017

Ans. (a)

See the explanation of above question.

11. When did the Bhopal Gas tragedy happen?

- (a) 2-3 Dec. 1984 (b) 2-3 Nov. 1984  
(c) 2-3 Dec. 1985 (d) 2-3 Nov. 1985

M.P.P.C.S. (Pre) 2013

Ans. (a)

See the explanation of above question.

12. From the decomposition of which of the following printing ink is formed :

- (a) Acetylene (b) Methane  
(c) Carbon tetrachloride (d) Benzene

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

Methane is colourless, tasteless and odourless gas which is found in the marshy area. Further, it is also present in the decomposition of organic matter and gas released by coal. It is used for the manufacture of carbon black. Carbon black is used for making black printing ink etc.

13. Methane gas producing field is –

- (a) Wheatfield (b) Paddy field  
(c) Cotton field (d) Groundnut field

U.P.P.C.S. (Pre) 2014

Ans. (b)

Paddy fields are a major source of atmospheric methane and have been estimated to contribute in the range of 50 to 100 million tonnes of the gas per annum.

14. Which of the following is correctly matched?

- | List-I                 | List - II      |
|------------------------|----------------|
| 1. Biogas              | Insecticide    |
| 2. Electrocardiography | Heart disorder |

3. D.D.T. Cow-dung

4. Nicotine Tobacco

Code :

- (a) 1, 2 and 4 (b) 2, 3 and 4  
(c) 2 and 3 (d) 2 and 4

U.P. Lower Sub. (Pre) 2002

Ans. (d)

In villages, we get gas for cooking and light from a biogas plant. From electrocardiography we detect the diseases related to the heart. D.D.T. is an insecticide. Nicotine is present in tobacco, which affects the health most.

15. Which one of the following chemicals helps in fruit ripening?

- (a) Ethephon (b) Atrazine  
(c) Isoproturan (d) Malathion

U.P. U.D.A./L.D.A. (Pre) 2010

U.P.P.C.S. (Pre) 2005

Ans. (a)

The Chemical Ethephon is often used in fruit ripening. Ethephon or ethrel is a commercial formulation which produce ethylene when dissolved in water at particular pH. This is a good substitute for calcium carbide.

16. Which of the following chemicals is used for ripening?

- (a) Sodium Chloride  
(b) Calcium Carbide  
(c) Potassium Chloride  
(d) None of the above

U.P.P.C.S. (Mains) 2011

Ans. (b)

In the artificial ripening of fruit, calcium carbide is sometimes used as a source of acetylene gas, which is a ripening agent similar to Ethylene. Its chemical formula is  $\text{CaC}_2$ .

17. Calcium carbide is used for artificial ripening of green fruits because it produces :

- (a) Methylene (b) Auxin  
(c) Acetylene (d) Florigen

R.A.S./R.T.S. (Pre) 2016

Ans. (c)

See the explanation of above question.

18. Fruit ripening is promoted by which of the following?

- (a) Carbon dioxide (b) Sulphur dioxide

(c) Nitrogen

(d) Ethylene

U.P.R.O./A.R.O (Pre) 2014

Ans. (d)

Ethylene gas is an important natural plant hormone, used to force the ripening of fruits.

19. Which of the following gas mixture is used in welding gas?

- (a) Acetylene and Hydrogen (b) Oxygen and Acetylene  
(c) Hydrogen and Oxygen (d) Hydrogen and Helium

Uttarakhand P.C.S. (Pre) 2005

Ans. (b)

In gas welding, the mixture of oxygen and acetylene are used.

20. Gases used in welding are :

- (a) Oxygen and hydrogen (b) Oxygen and nitrogen  
(c) Oxygen and acetylene (d) Hydrogen and acetylene  
(e) None of the above/More than one of the above

63<sup>rd</sup> B.P.S.C. (Pre) 2017

Ans. (c)

See the explanation of above question.

21. Consider the following statements about acetylene :

1. It is used in welding industry.
2. It is a raw material for preparing plastics.
3. It is easily obtained by mixing silicon carbide and water.

Of these statements –

- (a) 1 and 2 are correct (b) 1 and 3 are correct  
(c) 2 and 3 are correct (d) 1, 2 and 3 correct

I.A.S. (Pre) 1998

Ans. (a)

Acetylene is also called as Ethyne ( $\text{C}_2\text{H}_2$ ). It is used in welding. Chloroethene (Vinyl chloride) is produced by acetylene which is used as raw material in plastic industry. The mixture of calcium carbide (not silicon carbide) and water produces acetylene.

22. For benzene which of the following statements is true?

- (a) It has six sigma and six pi bonds  
(b) It has twelve sigma and six pi bonds  
(c) It has twelve sigma and three pi bonds  
(d) It has six sigma and three pi bonds  
(e) None of the above.

Chhattisgarh P.C.S. (Pre) 2014

Ans. (c)